

GSK3 β -mediated phosphorylation of the microtubule-associated protein 2C (MAP2C) prevents microtubule bundling

Carlos Sánchez¹⁾²⁾, Mar Pérez²⁾, Jesús Avila

Centro de Biología Molecular “Severo Ochoa”, Facultad de Ciencias, Universidad Autónoma de Madrid, Cantoblanco, Madrid/Spain

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A major determinant of neuronal morphology is the cytoskeleton. And one of the main regulatory mechanisms of cytoskeletal proteins is the modification of their phosphorylation state via changes in the relative activities of protein kinases and phosphatases in neurons. In particular, the microtubule-associated protein 2 (MAP2) family of proteins are abundant cytoskeletal components predominantly expressed in neurons and have been found to be substrates for most of protein kinases and phosphatases present in neurons, including glycogen-synthase kinase 3 (GSK3). It has been suggested that changes in GSK3-mediated MAP phosphorylation may modify MT stability and could control neuronal development. We have previously shown that MAP2 is phosphorylated *in vitro* and *in situ* by GSK3 at Thr¹⁶²⁰ and Thr¹⁶²³, located in the proline-rich region of MAP2 and recognized by antibody 305. However, the function of the phosphorylation of this site of MAP2 is still unknown. In this study, non-neuronal COS-1 cells have been co-transfected with cDNAs encoding MAP2C and either wild type or mutated GSK3 β to analyze possible effects on microtubule stability and on the association of MAP2 with microtubules. We have found that GSK3 β phosphorylates MAP2C in co-transfected cells. Moreover, this phosphorylation is inhibited by the specific GSK3 inhibitor lithium chloride. Additionally, the formation of microtubule bundles, which is observed after transfection with MAP2C, was decreased when MAP2C was co-transfected with GSK3 β wild type. Microtubule bundles were not

observed in cells expressing MAP2C phosphorylated at the site recognized by antibody 305. The absence of microtubule bundles was reverted after treatment of MAP2C/GSK3 β wild type transfected cells with lithium chloride. Highly phosphorylated MAP2C species, which were phosphorylated at the site recognized by antibody 305, appeared in cells co-transfected with MAP2C and GSK3 β wild type. Interestingly, these MAP2C species were enriched in cytoskeleton-unbound protein preparations. These data suggests that GSK3-mediated phosphorylation of MAP2 may modify its binding to microtubules and regulate microtubule stability.

Abbreviations. GSK3 Glycogen-synthase kinase 3. – GSK3mut Glycogen-synthase kinase 3 mutated. – GSK3wt. – Glycogen-synthase kinase 3 wild type. – HMWMAP2 High-molecular weight MAP2 isoforms. – LMWMAP2 Low-molecular weight MAP2 isoforms. – MAP Microtubule-associated protein. – MAP2-P MAP2 phosphorylated at the site recognized by antibody 305. – MF Microfilament. – MT Microtubule. – PBS Phosphate-buffered saline.

Introduction

Microtubule-associated protein 2 (MAP2) is a family of heat-stable proteins very abundant in the mammalian central nervous system, where they may have an important role in neurite outgrowth and neuronal plasticity (Dayhoff et al., 1994; Woolf, 1998). The multiple human MAP2 isoforms expressed in neurons result from the alternative splicing of 20 exons from a single gene located in chromosome 2 (Shafit-Zagardo and Kalcheva, 1998). Traditionally, MAP2 isoforms have been divided in two groups: *a*) High-molecular weight MAP2 (HMWMAP2), which includes MAP2A and MAP2B, with molecular masses of 280 and 270 kDa, respectively, and *b*) Low-molecular weight MAP2 (LMWMAP2), which includes MAP2C and MAP2D, with molecular masses of 70 and

¹⁾ Dr. Carlos Sánchez, Centro de Biología Molecular “Severo Ochoa”, Facultad de Ciencias, Universidad Autónoma de Madrid, Cantoblanco, E-28049 Madrid/Spain, e-mail: csanchez@cbm.uam.es, Fax: 34–913974799.

²⁾ Both authors have equally contributed to this work.

75 kDa, respectively. LMWMAP2 contains the N- and C-terminal regions of HMWMAP2 linked together and lacks a long central projection domain. Both groups of isoforms have either three or four repeated sequences of 18 amino acids, spaced by 13–14 amino acids, that constitute the tubulin-binding domain at the C-terminal region of MAP2. All MAP2 isoforms have a proline-rich region just before the tubulin-binding domain which presumably regulates its function (Doll et al., 1993; Ferralli et al., 1994; Ludin et al., 1996). Both the tubulin-binding and proline-rich regions are relatively well conserved in other related MAPs such as tau and MAP4 (Gustke et al., 1994; Goode et al., 1997; Preuss et al., 1997; Biernat and Mandelkow, 1999).

MAP2 has been implicated in neuronal outgrowth and polarity by several studies in neuronal and non-neuronal cells (Avila et al., 1993; Craig and Banker, 1994; Hirokawa, 1994; Matus, 1994; Heidemann, 1996; Letourneau, 1996). Thus, neurite growth is inhibited when MAP2 expression is suppressed in neuronal cell cultures after treatment with specific antisense oligonucleotides (Dinsmore and Solomon, 1991; Caceres et al., 1992; Sharma et al., 1994). Additionally, the outgrowth of cytoplasmatic elongations, which are similar to neurites, is observed when MAP2 is overexpressed in cultured non-neuronal cells (Leclerc et al., 1993; Leclerc et al., 1996; Kalcheva et al., 1998; Boucher et al., 1999).

One of the most dramatic effects observed after transfection of non-neuronal cells with cDNAs encoding MAP2 is the formation of bundles of long and stiff MTs, which are usually observed crossing the cytoplasm or below the inner surface of the membrane (Umeyama et al., 1993; Burgin et al., 1994; Ferralli et al., 1994; Takemura et al., 1995; Kaech et al., 1996; Leclerc et al., 1996; Ludin et al., 1996; Kalcheva et al., 1998). However, the mechanism of MAP2-induced bundling is still not clear. One possibility is that bundling may be the consequence of the crosslinking of MTs by MAP2 dimerization. In support of this view, the electron microscopic observation of MT bundles in MAP2-transfected cells showed that MTs were crosslinked by some molecules. They may correspond to MAP2 dimers since the spacing between MTs was similar to the one found in axons when the cells were transfected with LMWMAP2, and similar to the one found in dendrites when the transfection was done with HMWMAP2 (Hirokawa, 1994; Takemura et al., 1995; Leclerc et al., 1996; Shafit-Zagardo and Kalcheva, 1998). Indeed, MAP2 lacking its C-terminal region, which does not form dimers in vitro (Wille et al., 1992), fails to induce MT bundles in transfected cells. Curiously, MAP2 constructs with more MT stiffness capacity have a greater ability to induce MT bundling in these transfected cells (Ferralli et al., 1994; Ludin et al., 1996; Felgner et al., 1997). Actually, long neurite-like cytoplasmatic processes filled with MT bundles are immediately formed in MAP2-transfected non-neuronal cells after destabilization of the cortical actin cytoskeleton with cytochalasin (Matus, 1994; Kaech et al., 1996; Boucher et al., 1999). Thus, MT bundling might simply result from the MAP2-mediated stabilization and stiffening of MTs.

One of the main mechanisms of regulation of the function of MAP2 is phosphorylation. MAP2 is highly phosphorylated in vivo and its phosphorylation state is developmentally regulated (Schulman, 1984; Goto et al., 1985; Hernandez et al., 1987; Yamamoto et al., 1988; Hoshi et al., 1992; Ainsztein and Purich, 1994; Berling et al., 1994; Sánchez et al., 1995; Sánchez et al., 1996; Itoh et al., 1997; Sánchez et al., 1998). The MAP2

phosphorylation state may regulate MT dynamics (Tsuyama et al., 1987; Brugg and Matus, 1991; Ainsztein and Purich, 1994; Itoh et al., 1997; Wang et al., 1998), organelle transport (Sheetz et al., 1998), susceptibility of MAP2 to proteolysis (Alexa et al., 1996) and neuronal plasticity (Díaz-Nido et al., 1993; Dayhoff et al., 1994; Mandell and Banker, 1996a; Quinlan and Halpain, 1996; Philpot et al., 1997; Sánchez et al., 1997; Woolf, 1998).

One of the protein kinases that phosphorylates MAP2 is glycogen-synthase kinase 3 (GSK3). Two mammalian GSK3 isoforms encoded by a distinct gene have been cloned and named α (51 kDa) and β (47 kDa) (Wodarz and Nusse, 1998). GSK3 is highly concentrated in brain (Woodgett, 1990; Takahashi et al., 1994), where it was initially identified as tau-protein kinase I (TPKI) (Ishiguro, 1997). It has been demonstrated that GSK3 phosphorylates in vitro Ser¹³⁶ on MAP2, located next to the RII-binding domain and recognized by antibody AP18 (Berling et al., 1994). On the other hand, we have shown that MAP2 is efficiently phosphorylated in vitro by GSK3, characterizing the phosphorylation of the Thr¹⁶²⁰ and Thr¹⁶²³, which are located in the proline-rich region and are recognized by antibody 305 (Sánchez et al., 1996). Furthermore, we have observed that this MAP2 sequence is dephosphorylated after the treatment of cultured cerebellar granule neurons with lithium, which is a GSK3 inhibitor (Sánchez et al., 1998). However, it is still unknown how GSK3-mediated phosphorylation may regulate the function of MAP2. Hence, we have studied the phosphorylation of MAP2 by GSK3 β in co-transfected COS-1 cells to analyze possible effects on MT organization.

Materials and methods

Materials

Polyclonal antibody 305 recognizes the sequence RTPGTGTPSY (comprising residues 1616–1626 of MAP2 according to the sequence published by Lewis et al., 1988) phosphorylated at Thr¹⁶²⁰ and/or Thr¹⁶²³, in HMWMAP2 and LMWMAP2 isoforms (Sánchez et al., 1996) (1:5000 in Western blot and 1:400 in immunofluorescence). The monoclonal antibody HM2, which was purchased from Sigma, recognized both HMWMAP2 and LMWMAP2 isoforms (1:10000 in Western blot and 1:50 in immunofluorescence). Additionally, α -tubulin antibody was purchased from Amersham Corp (United Kingdom) (1:10000 in Western blot and 1:500 in immunofluorescence); β_2 -tubulin was recognized by the rabbit polyclonal antibody Ab196, previously characterized by Armas-Portela et al. (1999) (1:200 in immunofluorescence) and the antibodies raised against tyrosinated and acetylated tubulin were purchased from Sigma (1:5000 in Western blot). Finally, the antibody against GSK-3 β was purchased from Transduction Laboratories (Lexington, KY) (1:1000 in Western blot and 1:500 in immunofluorescence). LiCl, used as a specific inhibitor of GSK-3 β , was purchased from Sigma (St. Louis, MO/USA).

Plasmids

The plasmid pCMVMAP2Cneomyc contains a 1750-bp fragment of the cDNA encoding the rat MAP2C, whose expression is directed by the CMV promoter. This cDNA was subcloned into NotI-digested pCMV vector (Nicole Leclerc, Kosik Laboratory). The cDNAs encoding wild-type GSK-3 β from *Xenopus laevis* and the K85R mutant GSK-3 β were subcloned into the plasmid pcDNA3 (Invitrogen) (Domínguez et al., 1995).

Cell culture and transfection studies

African green monkey kidney fibroblast (COS-1) cells were grown in Dulbecco's modified Eagle's medium supplemented with 5% fetal calf serum. Cells were transfected by the use of lipofectin (Gibco-BRL) according to the manufacturer's instructions. Unless stated, cells were treated with LiCl (20 mM for 3 h) 48 hours after transfection.

Protein extracts, gel electrophoresis and Western blot

Cells were harvested in chilled phosphate-buffered saline (PBS), resuspended and homogenized in buffer containing 50 mM HEPES, pH 7.4, 10 mM EDTA, 0.1% Triton X-100, 100 mM NaF, 0.1 mM sodium orthovanadate, 1 mM phenylmethylsulfonyl fluoride, 10 µg/ml leupeptin, 10 µg/ml pepstatin and 10 µg/ml aprotinin. Lysates were centrifuged at 10000g for 10 min at 4°C and boiled for 5 min in Laemmli's buffer. Protein samples were quantitated by the BCA protein assay. SDS-PAGE was carried out using 10% or 8% gels which were subsequently transferred to nitrocellulose. Immunoreactive proteins were visualized by the use of peroxidase-conjugated anti-rabbit or anti-mouse antibodies (GIBCO) and enhanced by chemiluminescence detection (ECL kit from Pierce).

Cell fractionation

Cells were incubated with buffer PHEM (80 mM Pipes, pH 6.9, 1 mM MgCl₂, 5 mM EGTA, 0.5% Triton X-100) for 30 seconds. This time is probably brief enough to avoid alterations in cells during the extraction procedure. Supernatant (detergent-soluble fraction), with cytoskeleton-unbound proteins, was first harvested and detergent-insoluble components which correspond to the cytoskeleton-bound proteins, were subsequently resuspended in PHEM without Triton X-100.

Immunofluorescence microscopy

Cells plated on polylysine-coated coverslips were fixed with 4% paraformaldehyde for 30 min. After the incubation of coverslips with PBS supplemented with 0.1% Triton X-100 (PBS-TX), for 10 min, they were blocked with 1% fetal calf serum in PBS-TX for further 10 min. Incubation with primary antibodies was carried in PBS-TX for 45 min. Coverslips were rinsed three times with PBS-TX and incubated 30 min with Oregon green or Texas-Red-conjugated secondary antibodies (1:400) (Molecular Probes). Finally, they were rinsed with PBS-TX and mounted in Fluoromount. Coverslips were analyzed using a Zeiss epifluorescence microscope. Films were scanned in FilmScan 200 (EPSON), and images were processed in Adobe Photoshop 5.02 on a PC workstation. Quantification of cells was achieved by counting cells positive for antibody binding in 10 different fields with a ×40 objective, which comprised around 100 cells.

Results

MAP2C is phosphorylated by GSK3β in COS-1 transfected cells

To test if MAP2C was phosphorylated by GSK3β at the site recognized by antibody 305, Western blot analyses were performed using whole protein extracts from COS-1 cells co-transfected with cDNAs encoding MAP2C and either GSK3β wild type (GSK3wt) or an inactive mutated form of GSK3β (GSK3mut) (Fig. 1A). Additionally, densitometry of Western blots was performed. Values obtained were normalized for the reaction of antibody HM2 (MAP2) and antibody 305 (MAP2-P) to take into account the different amounts of whole MAP2 present in each sample (Fig. 1B). As a result, a 3.5-times increase in MAP2 phosphorylated at the site recognized by antibody 305 (MAP2-P) was obtained in MAP2C/GSK3wt co-transfected cells. Although a slight increase of MAP2-P may be observed in MAP2C/GSK3mut co-transfected cells (Fig. 1A),

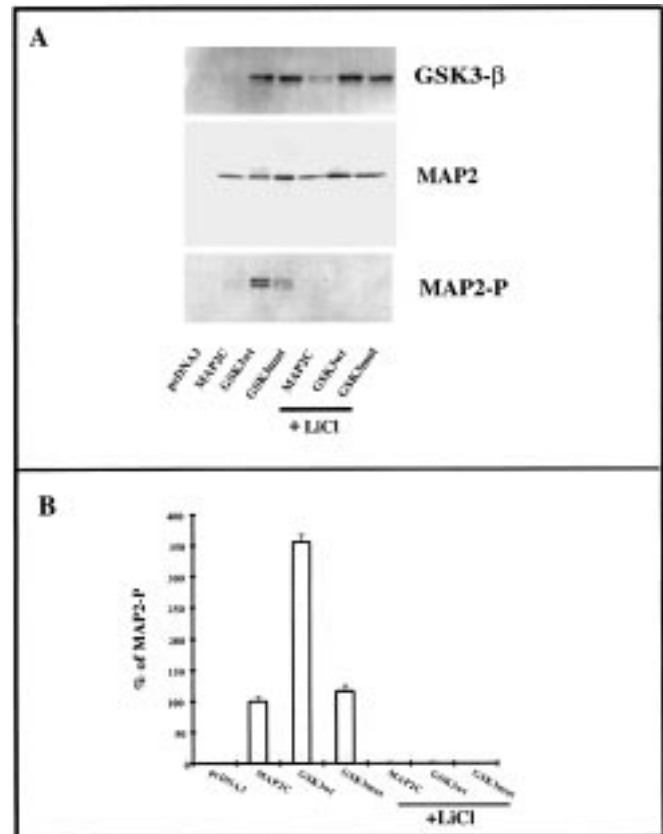


Fig. 1. MAP2C is phosphorylated by GSK3β in COS-1 transfected cells. **A)** COS-1 cells were transfected with plasmid without insert (pcDNA3), with the cDNA encoding MAP2C (MAP2C) or co-transfected with both MAP2C and either GSK3β wild type (GSK3wt) or GSK3β mutated (GSK3mut). In some cases, transfected cells were treated with the specific GSK3 inhibitor LiCl (20 mM for 3 hours), two days after transfection (LiCl). Whole protein extracts from these cells were analyzed by 10% SDS-PAGE and Western blots. GSK3β was detected by an antibody purchased from Transduction Laboratories, MAP2C by antibody HM2 (Sigma) and MAP2-P by antibody 305. **B)** Results of the densitometry of Western blots shown in panel A are indicated in this histogram (corresponding to three different experiments). The immunoreactivity of antibody 305 (MAP2-P) was normalized to the one of antibody HM2 (MAP2) to counteract the different amounts of MAP2C that may be present in samples (see also text). The values are indicated as percentage (%), where 100% corresponds to the amount of MAP2-P found in MAP2C-transfected cells. An increase in MAP2-P was observed in MAP2C/GSK3wt co-transfected cells, which was inhibited in the presence LiCl.

it may be due to a higher amount of MAP2 loaded in this sample, as indicated by normalized densitometry values (Fig. 1B). In addition, the increase in phosphorylation of MAP2C observed in MAP2C/GSK3wt co-transfected cells was inhibited after treatment with LiCl (Fig. 1), which is being extensively used as a specific GSK3 inhibitor (Klein and Melton, 1996; Stambolic et al., 1996; Hedgepeth et al., 1997; Hong et al., 1997; Muñoz-Montano et al., 1997; Cui et al., 1998).

Immunofluorescence analysis of MAP2C/GSK3wt co-transfected cells showed that 93 ± 5% of cells expressed both proteins, although only 50 ± 10% of cells expressing GSK3 also expressed MAP2-P (Fig. 2A,B). This latter proportion was decreased to 10 ± 7% in MAP2C/GSK3mut co-transfected cells (Fig. 2C,D).

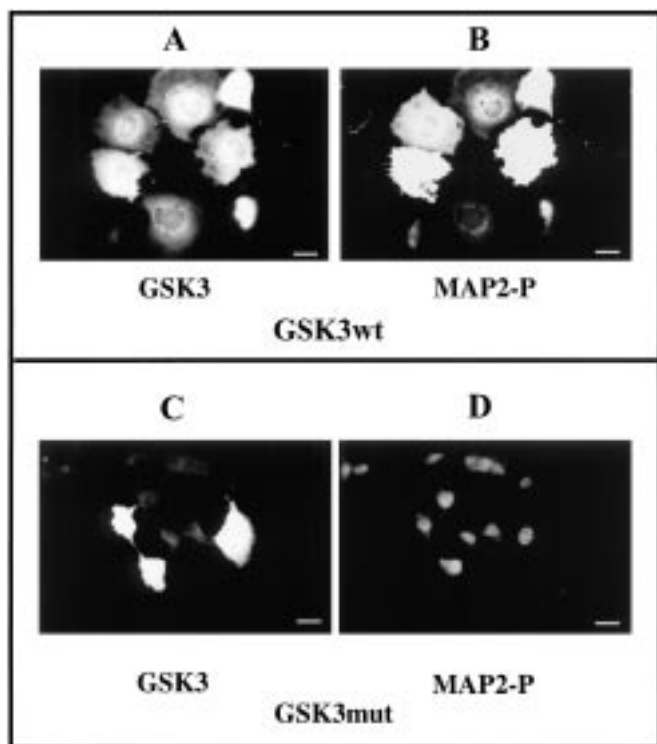


Fig. 2. Increase of MAP2-P in GSK3wt-transfected cells. Immunofluorescence of MAP2C/GSK3wt (GSK3wt) and MAP2C/GSK3mut (GSK3mut)-transfected cells fixed two days after transfection. GSK3 was recognized by an antibody against GSK3 β (Transductions Laboratories) and a secondary antibody coupled to Oregon green (A, C). MAP2-P was detected by antibody 305 and a secondary antibody coupled to Texas-Red (B, D). Most of the cells expressing GSK3 contained MAP2-P in MAP2C/GSK3wt-transfected cells, whereas no immunoreactivity with antibody 305 was found in MAP2C/GSK3mut-transfected cells. Bar, 10.8 μ m.

Some aspects must be considered to interpret these results. First, an endogenous GSK3 β activity subsists in COS-1 cells, triggering an endogenous MAP2C phosphorylation in the absence of exogenous GSK3 β . Actually, a slight increase in MAP2-P could be observed in MAP2C and MAP2C/GSK3mut co-transfected cells (Fig. 1). Second, GSK3wt used for these studies is not constitutively activated, thus it may be regulated during transfection of COS-1 cells. This might explain that only 50% of MAP2C/GSK3wt co-transfected cells expressing GSK3 contained MAP2-P (Fig. 2). And third, only 10% of COS-1 cells achieved the highest level of transfection efficiency in our culture conditions, although it is possible to find cells with lower degrees of transfected protein expression. This point should be taken into account to explain the results of Western blot analyses using whole cell extracts.

MT bundling is decreased after GSK3 β -mediated phosphorylation of MAP2C

Overexpression of MAP2 in non-neuronal cells induces the rearrangement of MTs, leading to the formation of long bundles of MTs that are capable of inducing stiff processes in these cells (Umeyama et al., 1993; Ferralli et al., 1994; Takemura et al., 1995; Kaech et al., 1996; Leclerc et al., 1996; Kalcheva et al., 1998). We observed that $80 \pm 10\%$ of MAP2C-

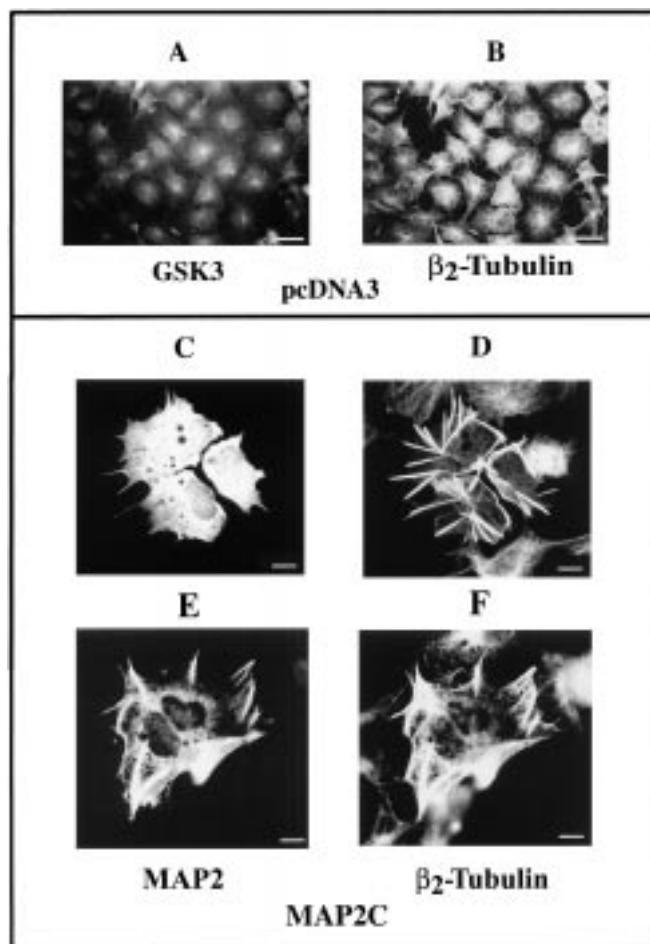


Fig. 3. MAP2C transfection induces MT bundles. COS-1 cells were transfected with plasmid without insert (pcDNA3) (A, B) or with the cDNA encoding MAP2C (MAP2C) (C, D). MAP2C was detected by antibody HM2 (Sigma) (MAP2), tubulin was recognized by antibody Ab196 against β_2 -tubulin (β_2 -Tubulin) and GSK3 by a GSK3 β antibody (Transductions Laboratories) (GSK3). Secondary antibodies were coupled to Oregon green for HM2 and GSK3 β antibodies and Texas-Red for β_2 -tubulin antibody. Immunofluorescence of MAP2C-transfected cells shows the formation of MT bundles in cells expressing MAP2C (C, D). Association of MAP2C with MTs could be observed with lower levels of MAP2C expression (E, F). In contrast, MT bundles are not observed in the absence of MAP2C or GSK3 expression (A, B). Bar (A, B), 10.8 μ m. Bar (D, E), 7.1 μ m. Bar (E, F), 6.5 μ m.

transfected cells showed MT bundles (Fig. 3C,D) (MAP2C), while no MT bundles were formed in the absence of MAP2C or GSK3 (Fig. 3A,B) (pcDNA3). Furthermore, when the level of MAP2 expression was lower it was possible to observe an association of MAP2C with MTs (Fig. 3E,F). Similarly, when COS-1 cells were co-transfected with both MAP2C and GSK3mut, MT bundles appeared in $80 \pm 7\%$ of the cells (Fig. 4E,F). In contrast, an inhibition of MT bundling could be detected in MAP2C/GSK3wt co-transfected cells (Fig. 4A–D). In this case, MT bundles were only found in $50 \pm 12\%$ of co-transfected cells. This MT bundling inhibition was not observed when cells were co-transfected with both MAP2C and GSK3mut (Fig. 4E,F). A decrease in MT bundling after transfection with GSK3 has been previously suggested (Lovestone et al., 1996; Wagner et al., 1996).

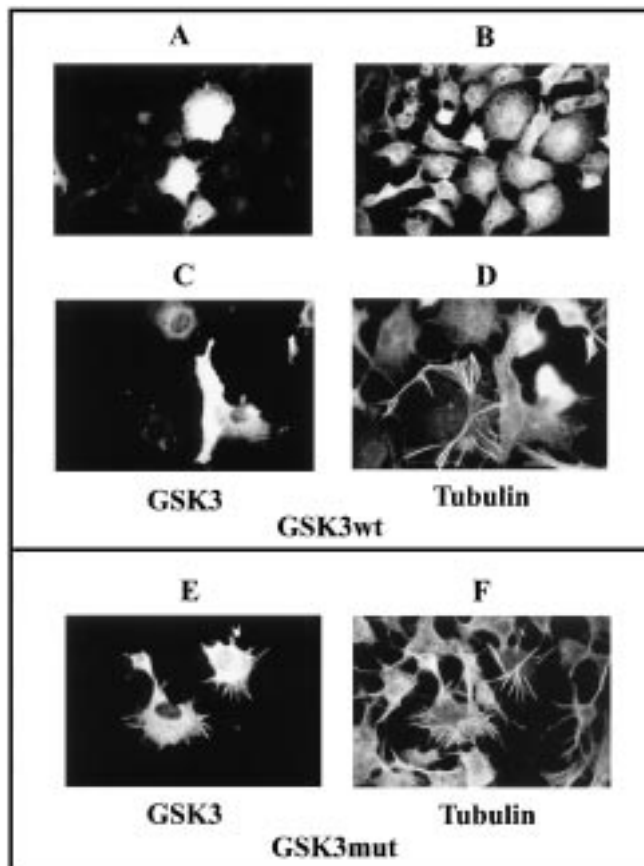


Fig. 4. Expression of GSK3wt decreases MT bundling. Immunofluorescence of MAP2C/GSK3wt (GSK3wt) or MAP2C/GSK3mut (GSK3mut) co-transfected cells. GSK3 was detected by an antibody against GSK3 β (Transductions Laboratories) (A, C, E) and tubulin was recognized by antibody Ab196 against β_2 -tubulin (B, D, F). Secondary antibodies were coupled to Oregon green for GSK3 β antibody and Texas-Red for β_2 -tubulin antibody. MT bundling was not observed in MAP2C/GSK3wt co-transfected cells (A, B). In particular, notice in panels C and D that the cell showing a higher GSK3 β expression did not contain MT bundles, whereas the adjacent cell with undetectable levels of GSK3 β did. In contrast, MAP2C/GSK3mut-transfected cells expressing GSK3 β showed MT bundles (E, F).

Since GSK3 phosphorylates MAP2 at the sequence recognized by antibody 305, we tested whether this modification of MAP2 might alter MT bundling in COS-1 cells. MT bundles were observed in only $25 \pm 10\%$ of cells in MAP2C/GSK3wt co-transfections, showing the expression of MAP2-P (Fig. 5 A,B). Conversely, MT bundles were detected in $85 \pm 10\%$ of cells in MAP2C/GSK3mut co-transfections. No expression of MAP2-P was found in this case (Fig. 5C,D). Furthermore, when MAP2C/GSK3wt co-transfected cells were treated with LiCl, cells expressing GSK3 started to show MT bundles (Fig. 6C,D), which may be correlated with the decrease of MAP2-P observed (Fig. 6A,B). Thus, the GSK3 β -mediated phosphorylation of MAP2C, at least at the sequence recognized by antibody 305, may be implicated in MT bundling.

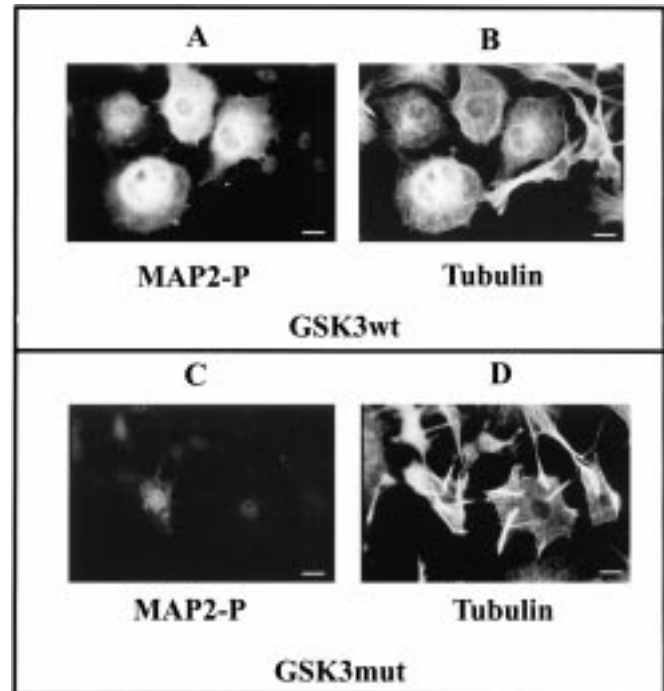


Fig. 5. MT bundling inhibition in cells expressing MAP2-P. COS-1 cells were co-transfected with MAP2C and either GSK3wt (GSK3wt) or GSK3mut (GSK3mut). MAP2-P was recognized by antibody 305 and a secondary antibody coupled with Texas-Red (A, C). Tubulin was tested with an antibody against α -tubulin (Sigma) and a secondary antibody coupled to Oregon green (B, D). Note the absence of MT bundles in MAP2C/GSK3wt co-transfected cells, which have a high level of expression of MAP2-P (GSK3wt). In contrast, MAP2C/GSK3mut co-transfected cells, which do not present MAP2-P, contained MT bundles. Bar, 7.7 μ m.

Highly phosphorylated MAP2C species with lower affinity for the cytoskeleton are generated in GSK3wt-transfected cells

In order to evaluate the effect of MAP2C phosphorylation by GSK3 β on its association with MTs, detergent-insoluble and soluble fractions, which contain cytoskeleton-bound and unbound proteins, respectively, were isolated and blotted with several antibodies. Detergent-insoluble fractions contained polymerized MTs as indicated by the higher presence of acetylated and a lower proportion of tyrosinated tubulin (Fig. 7A) (MacRae, 1997). It can be compared with the presence of the same amounts of total α -tubulin in both fractions (Fig. 7A). Western blots showed at least four MAP2C species with altered electrophoretic mobility, which may be a consequence of their different phosphorylation states (Fig. 7B). Electrophoretic patterns obtained were similar in MAP2C and MAP2C/GSK3mut co-transfected cells, but they differed from the one found in MAP2C/GSK3wt co-transfected cells. The higher electrophoretic mobility MAP2C species (band 1) (Fig. 7B) were not recognized by antibody 305 (MAP2-P), which indicates a low phosphorylated state of MAP2C at least at the sequence recognized by antibody 305. These species of MAP2C were present in both soluble and insoluble fractions (MAP2). The second group of MAP2C species (band 2) was specifically induced in MAP2C/GSK3wt-transfected cells. Interestingly, 60% of these MAP2C species were present in the cytoskeleton-unbound

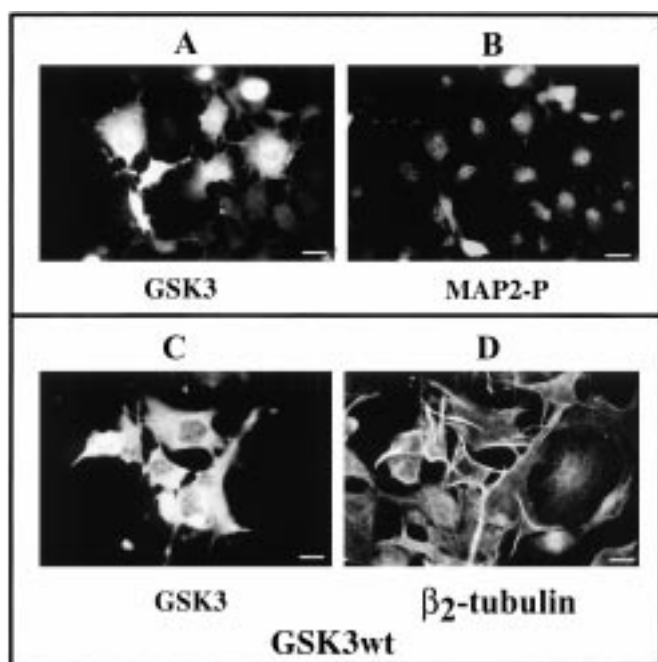


Fig. 6. MAP2C/GSK3wt co-transfected cells show MT bundles after LiCl treatment. MAP2C/GSK3wt co-transfected COS-1 cells were treated with the specific GSK3 inhibitor LiCl (20 mM for 3 hours) after two days of transfection. Then, cells were fixed and double immunostained with an antibody against GSK3 β (Transductions Laboratories) (GSK3) and antibody 305 (MAP2-P) or antibody Ab196 against β_2 -tubulin (β_2 -tubulin). GSK3 antibody was developed by a secondary antibody coupled with Oregon green, whereas antibodies 305 and Ab196 were detected by a secondary antibody coupled with Texas-Red. After LiCl treatment, MAP2-P was not found in cells expressing GSK3 β , which then started to show MT bundles. Bar (GSK3/MAP2-P), 10.8 μ m. Bar (GSK3/ β_2 -tubulin), 7.7 μ m.

fraction. A third group of MAP2C species (band 3) was present in MAP2C, MAP2C/GSK3wt and MAP2C/GSK3mut co-transfected cells, with 50%, 70% and 60% of MAP2C, respectively, present in cytoskeleton-unbound fractions (MAP2). Both, groups 2 and 3 of MAP2C species, were phosphorylated at least at the sequence recognized by antibody 305 (Fig. 7B) (MAP2-P). Finally, the slower migrating MAP2C species (band 4), which may correspond to a higher phosphorylated state, were mainly induced in MAP2C/GSK3wt co-transfected cells (MAP2). These MAP2C species showed a strong immunoreactivity with antibody 305 (MAP2-P). Furthermore, 90% of whole MAP2C was present in cytoskeleton-unbound fractions (MAP2).

Discussion

Several conclusions can be obtained from the results described in this work: 1) MAP2C is an *in situ* substrate of GSK3 β , which phosphorylates at least the site recognized by antibody 305 on MAP2; 2) GSK3 β -mediated phosphorylation of MAP2C decreases MT bundling; and 3) GSK3 β -mediated phosphorylation of MAP2C results in highly phosphorylated MAP2C species with a lower affinity for MTs. These features will be discussed in more detail in the next sections.

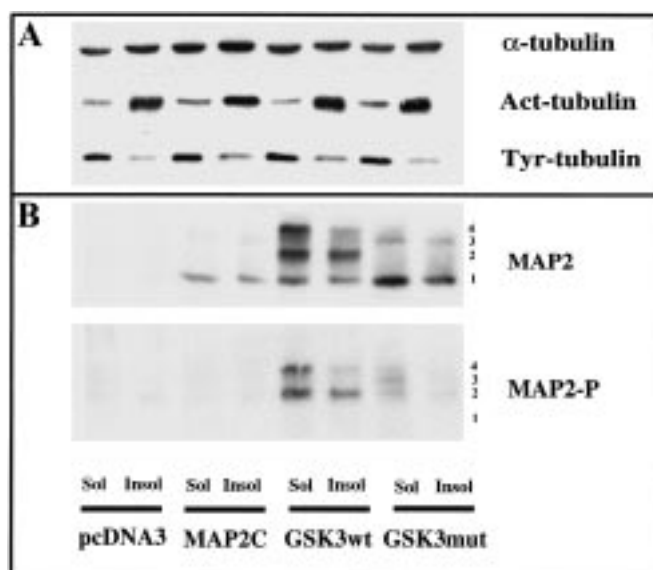


Fig. 7. GSK3 β -mediated phosphorylation of MAP2C decreases its association with the cytoskeleton. COS-1 cells were transfected with plasmid without insert (pcDNA3), with the cDNA encoding MAP2C (MAP2C) or co-transfected with both MAP2C and either GSK3 β wild type (GSK3wt) or GSK3 β mutated (GSK3mut). After two days of transfection, cells were incubated 30 seconds with extraction buffer containing 0.5% Triton X-100 (see Materials and methods for details). Detergent-soluble (Sol) and insoluble (Insol) fractions, which contain cytoskeleton-unbound and bound proteins, respectively, were isolated and blotted with antibodies that recognize α -tubulin, acetylated tubulin (Act-tubulin) and tyrosinated tubulin (Tyr-tubulin). MAP2C was recognized by antibody HM2 (MAP2) and phosphorylated MAP2C was detected by antibody 305 (MAP2-P). A high proportion of acetylated tubulin and a low proportion of tyrosinated tubulin were present in insoluble fractions (A), indicating that they mainly contain polymerized MTs. On the other hand, at least four MAP2C species were found in a Western blot after 8% SDS-PAGE (numbered from 1 to 4) (B). MAP2C species with higher electrophoretic mobility (band 1) (MAP2) may correspond to a lower phosphorylated state since they were not recognized by antibody 305 (band 1) (MAP2-P). In contrast, MAP2C species corresponding to band 4 had a lower electrophoretic mobility and were strongly recognized by antibody 305, which may indicate their higher phosphorylation state. These latter species of MAP2C were mainly present in the cytoskeletal unbound fraction (Sol) (see text for details).

GSK3 β phosphorylates MAP2 *in situ* at least at the site recognized by antibody 305

We recently demonstrated that MAP2 was phosphorylated *in vitro* by GSK3 at Thr¹⁶²⁰ and Thr¹⁶²³, which are located in the proline-rich region and are recognized by antibody 305 (Sánchez et al., 1996). Additionally, we found that lithium treatment of cultured cerebellar granule neurons inhibited MAP2 phosphorylation at this sequence, thus suggesting that MAP2 might be a substrate of GSK3 *in situ* (Sánchez et al., 1997). Present work seems to confirm these previous observations, since we have found a significant increase of MAP2-P in MAP2C/GSK3wt co-transfected COS-1 cells, which was inhibited by the treatment of transfected cells with lithium. This may suggest that MAP2 is indeed a good substrate for GSK3 β . Another additional site on MAP2 which is phosphorylated by GSK3 has been characterized (Berling et al., 1994). MAP2 was found to be phosphorylated at Ser¹³⁶, located next to the RII-binding domain and recognized by antibody AP18.

Interestingly, the site recognized by antibody 305 is conserved in tau protein, where it is recognized by antibodies tau-1 and AT-8 and can also be phosphorylated by PDPKs (Muñoz-Montañó et al., 1997). Many studies have emphasized that tau protein extensively phosphorylated by PDPKs could neither bind MTs nor promote MT assembly efficiently (Lovestone et al., 1996; Mandell and Banker, 1996b; Wagner et al., 1996; Goode et al., 1997; Hong et al., 1997; Preuss et al., 1997). Recent experiments performed in the insect Sf9 cells infected with baculoviruses encoding cDNA tau have shown that more than 80% of the total phosphate is incorporated in the Ser/Thr-Pro motifs of tau (Biernat and Mandelkow, 1999). Moreover, mutations abolishing phosphorylation of these motifs (from Ser/Thr to Ala) cause an increase in the formation of cytoplasmic extensions in tau-transfected cells. This suggests that the phosphorylation of Ser/Thr-Pro motifs hampers MT stabilization and process outgrowth (Biernat and Mandelkow, 1999). Thus, it may be possible that phosphorylation of MAP2 on this region could also modify its MT stabilization capacity.

MT bundling induced in MAP2C-transfected cells

When non-neuronal cells are transfected with MAP2, MT bundles with ten to one hundred times the diameter of one individual MT are induced (Lee and Brandt, 1992; Weisshaar et al., 1992; Leclerc et al., 1993; Umeyama et al., 1993; Matus, 1994). Actually, we have observed that more than 80% of COS-1 cells transfected with MAP2C contain MT bundles. It has been demonstrated that the third repeat of the tubulin-binding domain of MAP2 is essential for bundling, though the other repeats and the flanking regions, which includes the proline-rich region, have also an important contribution to this effect (Umeyama et al., 1993; Ferralli et al., 1994; Ludin et al., 1996). Furthermore, MAP2D, which is the LMWMAP2 isoform with four repeats in the tubulin-binding domain, induces MT bundles more efficiently than MAP2C, which has only three repeats. This suggests that the fourth repeat can improve the bundling efficiency of MAP2 (Ferhat et al., 1996).

Recent experiments have shown that only 50% of MAP2A-transfected COS-1 cells and 36% of MAP2B-transfected cells contained MT bundles (Kalcheva et al., 1998). Additionally, it has been demonstrated that MAP2B has less capacity than MAP2C to redistribute F-actin in the Sf9 insect cells (Boucher et al., 1999). These results may indicate that different MAP2 isoforms may have distinct capacities of stabilizing the cytoskeleton. This might be related with the different MT thickness, length and polarity that were found in cytoplasmic processes induced in the Sf9 insect cells when they were infected with baculoviruses encoding cDNAs either for MAP2C or HMWMAP2 (Leclerc et al., 1996). On the other hand, MAP2 phosphorylation may be also an important regulatory mechanism for its MT stabilizing activity.

Effects of GSK3 β -mediated phosphorylation of MAP2 on MT stability

It has been suggested that changes in GSK3-mediated MAP phosphorylation may modify MT stability and could control neuronal development (Lovestone et al., 1996; Wagner et al., 1996; Hong et al., 1997; Preuss et al., 1997; Biernat and Mandelkow, 1999; Muñoz-Montañó et al., 1999). We observed no change in different populations of tubulin in MAP2C/GSK3wt cotransfections, however we found a decrease of MT bundling in MAP2C/GSK3wt co-transfected cells. It should be

taken into account that only 10% of the whole population of cells analyzed by Western blot showed the highest level of expression of co-transfected proteins. Interestingly, the ratio of cells lacking MT bundles in MAP2C/GSK3wt co-transfections was similar to the proportion of cells expressing MAP2-P. Furthermore, the treatment of these transfected cells with the specific GSK3 β inhibitor LiCl reverted the decrease of MT bundling. These results suggest that GSK3 β may regulate MT bundling capacity of MAP2, and the site recognized by antibody 305 might be one of the phosphorylation sites of MAP2 involved.

On the other hand, the decrease of MT bundling observed in transfected cells expressing MAP2-P could be the consequence of a diminished association of MAP2 with MTs. Actually, a correlation has been found between the strength of the association of MAP2 with MTs and their stiffening and bundling (Ferralli et al., 1994; Felgner et al., 1997). Phosphorylation of MAP2C by GSK3 β in MAP2C/GSK3wt co-transfected cells triggered several MAPC species. Some of them might correspond to a high degree of phosphorylation, since they showed a lower electrophoretic mobility and a strong immunoreaction with antibody 305. Most MAP2C species were present in both the cytoskeleton-bound and unbound fractions. In contrast, the slower migrating MAP2C was present mainly in the cytoskeleton-unbound fraction. Brugg and Matus (1991) already showed that only MAP2 with a precise phosphorylation state could associate with MTs. Neither highly phosphorylated MAP2 nor highly dephosphorylated MAP2 could bind to tubulin after microinjections into cells (Brugg and Matus, 1991). However, the relevant phosphorylation sites were not analyzed in that study. Curiously, GSK3 β -mediated phosphorylation of the MAP2 site recognized by antibody AP18 had no effects on binding or bundling capacities of MAP2 (Berling et al., 1994). This supports the idea that the function of MAP2 may be mainly influenced by the phosphorylation of individual sites rather than the whole amount of phosphate incorporated into the molecule, although this latter parameter may be also important. HMWMAP2 and LMWMAP2 have around 327 and 88 Ser/Thr residues, respectively, that may be potentially phosphorylated. Furthermore, HMWMAP2 has a total of 43 Ser/Thr-Pro motifs while LMWMAP2 has a total of 19 of these motifs, which may be targets for PDPKs (Berling et al., 1994; Sánchez et al., 1996). Thus, the existence of many potential phosphorylation sites on MAP2 makes it very difficult to understand the specific functions of each phosphorylation site, since the phosphorylation state of flanking regions may also affect the phosphorylation of a particular site. It may be possible that the synchronized phosphorylation of several residues, more than the phosphorylation of an individual sequence, could be responsible for the alteration of specific MAP2 functions. Additionally, several functions can be regulated by MAP2 such as bundling, binding, nucleation and stabilization not only of MTs but also of MFs. Moreover, MAP2 could also anchor protein kinases, phosphatases and/or neurotransmitter receptors to the cytoskeleton in dendrites and dendritic spines (Rossum and Hanisch, 1999). Thus, phosphorylation of MAP2 at particular sites may influence functions of this protein that are not usually analyzed in the conventional binding assays. The complexity of the regulation of site-specific MAP2 phosphorylation is illustrated by the case of the site recognized by antibody 305. This site is mainly phosphorylated by GSK3 in developing neurons in culture, but they become also phos-

phorylated by ERKs when cultured neurons mature (Sánchez, et al., 1998). Accordingly, the phosphorylation of MAP2 at a particular site may depend not only on the substrate specificities of protein kinases but also on the subcellular compartmentalization and/or specific cytoskeletal anchoring of protein kinases during neuronal development.

Thus, the development of reagents such as antibodies that can discriminate between distinct phosphorylated species of a protein may be useful in this respect. The identification of all sites that become phosphorylated on MAP2 *in vivo* is a prerequisite to obtain these specific antibodies that would be required to study the regulation of MAP2 phosphorylation in response to distinct extracellular signals under different physiological and pathological situations (Tokuda and Hatase, 1998; Gundersen and Cook, 1999).

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